

Completing the Market - DeFi Perpetual Futures as Spanning Instruments for Traditional Assets

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Abstract

Traditional markets for equities, indices and commodities are closed roughly two thirds of the calendar week, yet the news process that drives their fundamentals never stops. This paper studies whether decentralised perpetual futures on Hyperliquid, PAXG and other RWAs which trade 24/7 complete the market in the temporal dimension whether they span states of the world (weekend and overnight news realisations) against which no traditional instrument can be traded. Using roughly 17,000 aligned hourly observations across eleven assets from December 2025 to July 2026, I find that perpetual prices track cash markets nearly one for one when both are open (pooled $\beta = 0.94$, $R^2 = 0.82$) and remain unbiased, informative forecasts when cash markets are shut overnight perpetual returns predict equity opening gaps with $\beta \approx 0.7$ – 1.0 and R^2 up to 0.54 , and pooled weekend directional accuracy reaches 77% overall and 89% for large moves. Vector error-correction estimates show cash prices adjusting toward perpetual prices, with Hasbrouck information shares of 0.73 – 0.83 for the perpetual leg in equities and natural gas. Critically, none of this constitutes alpha and signals are fully impounded by the open, out-of-sample trading nets negative Sharpe ratios after costs, and a reactive hedge placed fifteen minutes after weekend news captures little. The economic value is instead access a pre-emptive weekend hedge during the February 2026 Strait of Hormuz crisis recovered 91% of a gold position's gap loss (77% after 20bps costs) at funding costs measured in tens of basis points annualised. Efficiency and completeness are two sides of the same result the perps are valuable as hedging venues precisely because they offer no free lunch.

1. Introduction

An Arrow–Debreu market is complete when every state-contingent payoff can be replicated by traded instruments. The literature has mostly examined completeness across payoff states. Ross (1976) showed how options complete the market over terminal price states. This paper examines the neglected temporal dimension. Between Friday 21:00 UTC and Sunday 22:00 UTC, a portfolio manager holding gold, oil or S&P 500 exposure cannot trade the state “geopolitical shock realised on Saturday.” The claim exists economically the news arrives, the value of the position changes but no traditional instrument spans it. The market is incomplete not because a payoff is missing, but because a trading time is.

Decentralised finance has quietly filled this gap. Hyperliquid, an on-chain derivatives exchange, lists perpetual futures on traditional underlyings equity indices, single stocks, energy and metals that trade continuously. Whether these contracts genuinely complete the market, however, is an empirical question with two parts. First, fidelity that is do perpetual prices actually track the underlying, or do they trade as crypto beta in disguise? A hedge in an instrument that follows Bitcoin rather than Brent is worse than no hedge. Second, informativeness when the cash market is closed, is the perpetual price an unbiased forecast of where cash will reopen? If closure-hours prices were noise, the venue would offer trading without spanning.

I answer both questions and draw a sharp distinction that frames the whole paper the contribution of these instruments is completeness, not alpha. A recurring temptation in this literature is to interpret the predictive power of 24/7 prices over opening gaps as a trading signal. I show it is not the prediction is fully priced by the open, and every strategy built on it dies out of sample after realistic costs. That failure is not a weakness of the result it is the result. In a semi-strong efficient market, an instrument that lets you transact against weekend states should not also pay you for doing so. What it pays is an insurance service the ability to hedge when nothing else is open, at a cost measured in basis points of funding rather than in overnight gap risk measured in percent.

The paper relates to three strands of research. The price-discovery literature (Hasbrouck 1995; Gonzalo and Granger 1995) provides the machinery for asking where information is impounded when two venues trade the same asset; Kawaller, Koch and Koch (1987) established that futures lead cash intraday. The after-hours literature (Barclay and Hendershott 2003) shows that low-liquidity sessions still contribute disproportionate price discovery per trade. The crypto-market-structure literature (Makarov and Schoar 2020) documents both the arbitrage frictions and the increasing integration of on-chain venues. This paper joins them at their intersection showcasing on-chain venues performing price discovery for off-chain assets during the hours the off-chain market does not exist.

What are RWAs and Perp Contracts

RWAs are traditional financial assets represented on blockchains like equities, commodities, FX, indices, pre-IPO companies. On Hyperliquid these mostly exist not as tokenized ownership but as *synthetic price exposure* through perpetual futures deployed via HIP-3, the mechanism that lets any builder permissionlessly list a perp on any reference asset. So you get perps on WTI, Brent, gold, silver, S&P 500, Tesla, Nvidia, and even pre-IPO names on TradeXYZ, a platform built on Hyperliquid, rolled out pre-IPO perpetual futures tied to companies including SpaceX. The growth here has been dramatic: as of mid-2026, 23 of the top 30 assets on Hyperliquid by open interest are commodities and equities, not cryptocurrencies.

A perp is a futures contract with no expiry. Instead of converging to spot at settlement, it's tethered to the underlying's reference price via a *funding rate* which is when the perp trades above the index, longs pay shorts (and vice versa), typically every hour on HL. Traders post margin (USDC on Hyperliquid), take leveraged long/short positions, and positions are marked continuously with automated liquidation if margin falls below maintenance requirements. The funding mechanism replaces the delivery/settlement machinery of traditional futures i.e no physical delivery, no roll, no expiry calendar.

Hyperliquid is a purpose built layer-1 blockchain running a fully on-chain central limit order book (CLOB) not an AMM like Uniswap-style DEXs. Key mechanics:

- **On-chain matching:** Orders, cancels, matches, and liquidations all happen on-chain with sub-second finality via its HyperBFT consensus. It replicates a CEX experience (limit orders, depth, maker/taker fees) with DeFi's self-custody and transparency.
- **Self-clearing:** The protocol is its own clearinghouse. Margin, mark prices, funding, and liquidations are enforced by code, with the HLP (community liquidity vault) and insurance fund backstopping liquidations rather than a central counterparty.
- **Oracles:** Perps on RWAs reference external index prices (via oracle feeds), which is where the interesting weekend behavior arises. During traditional market closures the perp itself becomes the live price, since there's no cash market to anchor to.

PAXG and XAUT: tokenized gold backed by physical bullion

These are the two dominant tokenized gold products, and unlike Hyperliquid's synthetic perps, they are *fully collateralized claims on real metal*. Together they hold roughly nine tenths of the gold-backed token market.

PAXG (Pax Gold): Issued by Paxos Trust Company, each PAXG token represents one troy ounce of LBMA-approved gold stored in London vaults, with monthly third-party attestations and redeemability for physical gold. The backing is unusually granular and the gold sits in Brink's vaults in London, each PAXG token is linked to a specific serial-numbered gold bar that holders can verify through Paxos' allocation lookup tool, and KPMG performs the monthly attestations. Paxos now operates under a national trust charter overseen by the OCC, making it the compliance-grade choice for institutions.

XAUT (Tether Gold): Each XAUT token is backed by one troy ounce of 99.99% pure gold stored in Swiss vaults, with quarterly audits through BDO Italia, publicly available reserve reports, and on-chain verification of backing via the holder's Ethereum address. As of September 2025, Tether confirmed roughly 375,572 ounces (~11.6 tons) backing the token.

Institutions extending the model to silver, oil, and beyond

The gold blueprint is now being replicated across the commodity complex:

- **Silver:** Kinesis Silver (KAG) tokens each correspond to one ounce of investment-grade silver, fully backed by physical metal in insured, third-party-audited vaults. Binance launched XAG/USDT perps in January 2026 with 24-hour volume hitting \$1.32 billion, and Hyperliquid's SILVER-USDC pair has surged past \$1 billion in daily volume. Still early though tokenized gold carries about \$5.1 billion of market value against silver's roughly \$435 million.
- **WTI crude:** A startup called Energy Substantiation is attempting genuinely asset-backed oil a digital token (WTIC) tied to physical crude, where suppliers feed barrels into the system via reverse Dutch auction, monetizing dormant inventory like pipeline line fill and tank bottoms, holders can redeem at the daily spot close. It's working with about a dozen commodities firms including one major trading house. Tiny so far (~\$80k on-chain), but it's the physical-backing analogue to HL's synthetic WTI perp.



The RWAs such as PAXG and XAUT also present an interesting angle for this study as these RWAs can be redeemed for real assets and the companies operating them lie under proper jurisdictions. PAXG and XAUT are pegged as 1:1 basis and users have the option to redeem on chain tokens with real gold bars. Hence no matter how much token price deviates from gold users have the guarantee to get their money back.

2. Data and Methodology

The sample covers eleven Hyperliquid perpetuals with traditional underlyings SPX, Nasdaq (NDAQ), Nikkei 225, KOSPI 200, GOOGL, Brent, natural gas, silver, copper, platinum and palladium from first listing (December 2025 for the earliest contracts) to 8 July 2026, at hourly resolution roughly 2,200–5,000 hourly bars per asset and 17,071 aligned perp cash hourly return pairs in the pooled panel. Perpetual candles and funding-rate histories come from Hyperliquid's public API while cash and futures benchmarks come from exchange front-month futures and index data. Each hourly observation is classified into one of three sessions using the exchange calendar of its underlying (NYSE, CME, ICE, JPX, KRX) market open, after hours, and

weekend. A complementary two year daily and hourly dataset on PAXG (tokenised gold) versus spot gold provides a longer-history validation laboratory, since tokenised gold has traded 24/7 since 2020.

Fidelity is measured by regressing cash hourly log returns on contemporaneous perpetual returns, per asset and per session, and in a pooled panel with asset fixed effects. Closure-hours informativeness is measured by regressing the cash open-to-open gap on the perpetual's cumulative return over the corresponding closed window, with a strict signal cutoff before the underlying's opening auction (an audit of the NYSE 14:00 vs. 14:30 UTC boundary confirms zero post-open bleed bars). Price discovery is assessed with Johansen cointegration, a VECM whose adjustment coefficients identify which price error-corrects toward which, Hasbrouck (1995) information-share bounds, and Granger causality. Crypto-beta contamination is controlled with a two-factor regression adding contemporaneous Bitcoin returns, and for commodities a factor model including the dollar index. Validation uses 70/30 chronological splits, rolling-origin walk-forward folds with permutation p-values, BCa bootstrap intervals, Chow break tests, and 7-day rolling betas. Hedge economics are computed net of realised funding payments, oracle basis and execution-cost haircuts of 10–20bps.

3. Findings

3.1 Perpetual prices track cash nearly one-for-one

Table 1 reports the pooled panel. When both venues trade simultaneously, the hourly beta of cash on perpetual returns is 0.945 with an R^2 of 0.82 across 5,935 observations the fixed-effects and no fixed effects estimates are indistinguishable. Commodity contracts are the cleanest silver's market-open beta is 1.009 with $R^2 = 0.98$, platinum 0.993 ($R^2 = 0.95$), Brent 0.999 ($R^2 = 0.84$) and natural gas 1.015 ($R^2 = 0.90$). The weekend beta of 0.86 is estimated on only 222 hourly pairs (weekends supply few cash prints) and is noisier, but remains statistically indistinguishable from one and economically far from zero.

Table 1. Pooled regressions of cash hourly returns on Hyperliquid perpetual returns, by session (asset fixed effects Dec 2025 – Jul 2026, 11 assets).

Session	β (HL)	s.e.	t-stat	R^2	N
All sessions	0.924	0.017	55.5	0.675	17,071
Market open (RTH)	0.945	0.012	77.6	0.818	5,935
After hours	0.923	0.017	53.1	0.660	10,914
Weekend	0.863	0.256	3.37	0.425	222

The two-factor control addresses the crypto-beta concern directly adding Bitcoin returns leaves the perpetual beta essentially unchanged for every asset, with Bitcoin loadings between -0.22 and $+0.15$ and typically below 0.10 in absolute value. The commodity factor model tells the same story dollar-index betas are significant everywhere while Bitcoin exposure, where present, is small. These contracts price their underlyings, not the crypto cycle. The longer PAXG gold history confirms the mechanism over two years the pair is cointegrated (Engle–Granger $p < 0.001$) with a hedge ratio of 1.003, and the apparent daily beta of 0.75 in naive close-to-close data is a sampling-clock artifact sampling both legs at the same UTC hour restores $\beta = 0.92$, and the Dimson aggregated beta of 0.96 shows full economic exposure with the residual attributable purely to non-synchronous closes.

3.2 When cash is closed, the perpetual is the price

The predictive regressions establish that closure-hours perpetual prices are unbiased forecasts of the cash reopen. Overnight perpetual returns predict opening gaps with $\beta = 0.71$ ($R^2 = 0.46$) for SPX, $\beta = 0.80$ ($R^2 = 0.53$) for Nasdaq and $\beta = 1.02$ ($R^2 = 0.54$) for GOOGL, with directional accuracy of 76–81%. Pooled across all eleven assets, weekend directional accuracy is 76.8% over 246 asset-weekends, rising to 88.7% for moves above the 75th percentile of magnitude and 88% above the 90th precisely the tail states where a hedge matters.

The VECM results locate price discovery. For every equity contract and for natural gas, the perpetual's adjustment coefficient is statistically indistinguishable from zero while the cash leg error-corrects toward the perpetual the on-chain price anchors and the exchange price converges to it. Hasbrouck information-share midpoints for the perpetual leg are 0.83 (SPX), 0.82 (Nasdaq), 0.80 (GOOGL), 0.78 (Nikkei) and 0.73 (natural gas), with Granger causality from perpetual to underlying significant at the 1% level for the equity contracts. This echoes Barclay and Hendershott's (2003) finding that thin sessions can carry outsized information extended here to sessions in which the reference market does not merely thin out but ceases to exist.

3.3 The efficiency result: there is no alpha

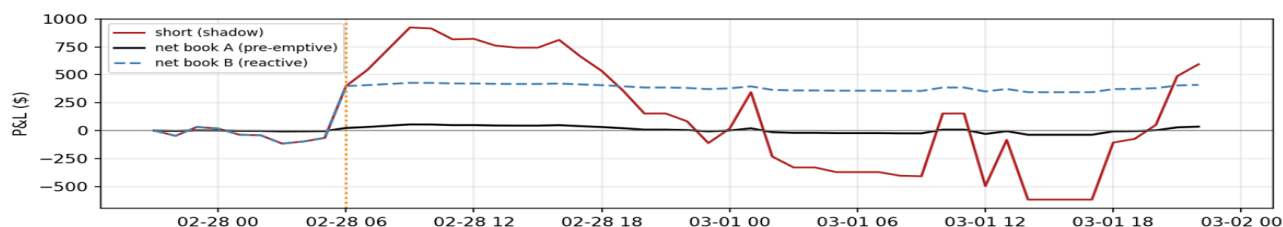
If gap prediction of this quality were tradeable, it would be an anomaly. It is not. A directional strategy trading the cash reopen on the sign of the 24/7 weekend move tested on the six-year PAXG–gold history to maximise statistical power earns a gross annualised Sharpe of 1.59 that collapses to -0.69 once the reopen haircut and realistic costs are applied which means that the gap the signal predicts is exactly the gap at which the trade executes. Walk-forward tests on the Hyperliquid panel show the same pattern betas are highly significant in-sample (permutation $p < 0.01$ for eight of eleven assets, positive 70/30 out-of-sample R^2 for ten of eleven), yet no specification produces a positive net out-of-sample Sharpe at any post-open horizon. A related-venue arbitrage between two tokenised golds survives frictionless backtesting (net Sharpe 2.28) but degrades to 0.57 under the joint stress of one-hour execution lag, realistic per-leg costs and one-sided shortability noise reversion, not economics. The market is semi-strong efficient with respect to its own 24/7 prices everything the perpetual knows on Sunday night is in the opening print on Monday.

3.4 What completeness is worth: hedge effectiveness net of costs

The value of the venue therefore lies entirely in transacting before the open and the February 27 – March 2, 2026 weekend provides a natural experiment. Strikes on Iran and the threatened closure of the Strait of Hormuz broke on Saturday with CME and ICE shut gold reopened $+2.22\%$ and WTI $+6.79\%$. Table 2 reports the hedge accounting for a short 1000z gold position (\$525k notional) and a short 1,000bbl WTI position (\$67k notional), hedged long in the corresponding Hyperliquid perpetual, net of realised funding ($+0.13\%/49h$ for gold, $+0.40\%/49h$ for WTI) and at execution haircuts of 0–20bps. These positions are at 15 minutes after the news hit the market, hence we can see WTI already risen due to the information flow.

Asset	Hedge scenario	Cash gap	HL return	Unhedged P&L	Gap recovered (net)
Gold	Pre-emptive (Fri close)	$+2.22\%$	$+2.25\%$	$-\$11,630$	90.9% (77.3% @20bps)
Gold	Reactive (news +15 min)	$+2.22\%$	$+1.47\%$	$-\$11,630$	56.3% (42.7% @20bps)
WTI	Pre-emptive (Fri close)	$+6.79\%$	$+4.63\%$	$-\$4,570$	60.8% (56.3% @20bps)
WTI	Reactive (news +15 min)	$+6.79\%$	$+0.00\%$	$-\$4,570$	-5.2%

Two lessons follow. First, the pre-emptive hedge placed at Friday's halt as a matter of policy rather than prediction recovers 91% of the gold gap (77% after 20bps costs) and 61% of the far more violent WTI gap. Second, the reactive hedge fails, and its failure is the efficiency result of Section 3.3 restated: fifteen minutes after the news, the perpetual had already moved, leaving -5% of the WTI gap recoverable. Completeness is an ex-ante property one must hold the spanning instrument before the state realises. Steady-state funding costs make the standing hedge affordable: across the panel, net annualised funding ranges from roughly -60 bps (Brent, natural gas, where oracle basis adds cost) to $+29$ bps (KOSPI), with the index contracts near zero (SPX -3 bps, Nasdaq -2 bps).



4. Robustness

The findings survive the standard objections. Boundary contamination reclassifying the ambiguous 14:00–14:30 UTC pre-open window leaves equity gap betas and directional accuracy essentially unchanged (SPX β moves from 0.71 to 0.77) with zero post-open bleed bars in either window. Structural stability shown by Chow

tests reject parameter constancy only for silver, 7-day rolling betas show no sign flips for any asset and standard deviations below 0.13. Crypto contamination shows the two-factor and dollar-factor controls of Section 3.1. Small weekend samples shows the weekend beta's wide standard error is acknowledged directly and is why the hedging conclusions lean on the event study and on the six-year tokenised-gold history, where weekend behaviour is estimated on hundreds of episodes. Sampling artifacts the PAXG Dimson correction shows how easily non-synchronous closes masquerade as economic under-reaction, a caution applied throughout the panel design. Finally, results are robust to fixed effects, to bootstrap inference on betas, and to the 10–20bps execution-cost haircuts in every hedge computation.

5. Discussion: Completing the Market in Time

Read together, the results describe a coherent economic object. During trading hours the perpetual is a near-perfect substitute for the cash asset ($\beta \approx 0.95$, $R^2 \approx 0.82$). During closure it becomes something the cash asset cannot be i.e a traded price for the untradeable state. That price is disciplined, cash error-corrects to it, its direction anticipates the reopen 77–89% of the time, and it embeds news within minutes yet it yields no exploitable excess return. In Ross's (1976) sense, options completed the market over payoff states without themselves being mispriced these contracts complete it over time states in exactly the same way. The absence of alpha is what certifies the venue a hedger can trust the Sunday-night perpetual price for the same reason a speculator cannot profit from it.

The practical implication is a shift in how institutions should think about weekend risk. Gap risk over closures is conventionally treated as unhedgeable and absorbed via VaR add-ons, reduced Friday positioning, or options bought during the week at implied volatilities that embed the closure premium. A standing pre-emptive perpetual hedge converts that lumpy, unpriceable exposure into a smooth carry cost of tens of basis points annualised plus basis risk that this paper quantifies session by session. The scorecard's asset-by-asset verdicts are the operationally useful output completeness is a property to be verified per contract, not assumed per venue.

6. Conclusion

DeFi perpetual futures on traditional assets are best understood not as a new casino but as the market's missing hours. Across eleven contracts and seven months of hourly data, they price their underlyings faithfully when both venues are open, lead price discovery when only they are open, forecast reopening prices without bias, while realised weekend gap losses by up to 91% for a hedger who is positioned before the news. Efficiency and completeness arrive together, as theory says they should. The research agenda this opens is concrete and that longer samples as contracts season, microstructure of the oracle basis under stress, capacity limits of on-chain liquidity against institutional hedge sizes, and the equilibrium question of whether traditional exchanges respond to competition from venues that never close. Along with that innovation of RWAs like PAXG which can be redeemed for real gold and the institutions providing these services are already under governance of SWISS and US Government showcase a huge unexplored area for RWAs as well. Similar to PAXG many firms are now working towards launching their own RWAs which can be redeemed with real assets mainly commodities could be the potential for an under discovered market.

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